

Unit 13: Logical Venn Diagram and Set Theory, Syllogism

CONTENTS

Objective

Introduction

11.1 Venn Diagrams

11.2 Set

11.3 Operation on Sets

11.4 Venn Diagram with Two Attributes

11.5 Venn Diagram with Three Attributes

11.6 Representation of Sets

11.7 Set Builder Form or rule Method

11.8 Finite and Infinite Sets

11.9 Empty Set or Null Set

11.10 Venn Diagram

11.11 SYLLOGISMS: Deductive Reasoning

Summary

Keywords

Self Assessment

Answer for Self Assessment

Review Questions

Further Readings

Objective

The key concepts explained in this unit, including area a learner is expected to learn and master after going through the unit are: -

- We will understand how to show the relationships among things or finite groups of things.
- We will understand what are sets
- We will understand what are different types of sets
- We will understand what are operations on a sets
- We will understand what is Syllogism.

Introduction

The concept of set is fundamental in all branches of mathematics. Sets are the most basic tools of mathematics which are extensively used in developing the foundations of relations and functions, logic theory, sequences and series, geometry, probability theory, etc. In fact, these days most of the concepts and results in mathematics are expressed in the set theoretic language.

The modern theory of sets was developed by the German mathematician Georg Cantor (1845–1918AD). In this chapter, we will study some basic definitions and operations involving sets. We will also discuss the applications of sets.

11.1 Venn Diagrams

The diagrams drawn to represent sets are called Venn – Euler diagrams or simply Venn diagrams. In Venn diagrams, the universal set U is represented by points within a rectangle and its subsets are represented by points in closed curves (usually circles) within the rectangle. If a set A is a subset of B , then the circle representing A is drawn inside the circle representing B . If A and B are not equal but they have some common elements, then to represent A and B we draw two intersecting circles. Two disjoint sets are represented by two non-intersecting circles.

11.2 Set

We observe that in nature, varieties of objects occur in groups. These groups are given different names such as, a collection of books, a bunch of keys, a herd of cattle, an aggregate of points, etc., depending on the characteristic of objects they represent. In literal sense, all these words have the same meaning. (i.e., a group or a collection). In mathematical language, we call this collection of objects, a set. From the above examples, it can be seen that each collection has a well-defined property (characteristic) of its own.

Thus, a set is a well-defined collection of objects. When we say well defined, we mean that the objects follow a given rule or rules. With the help of this rule, we will be able to say whether any given object belongs to this set or not. For example, if we say that we have a collection of short students in a class, this collection is not a set as 'short students', is not well defined. However, if we say that we have a collection of students whose height is less than 5 feet, then it represents a set.

It is not necessary that a set may consist of same type of objects, For example, a book, a cup and a plate lying on a table may also form a set, their common property being that they form a collection of objects lying on the table.



Example 1: Some other examples of sets are:

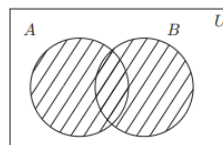
- (i) The set of numbers 1, 3, 5, 7, 9, 14.
- (ii) The set of vowels in the alphabets of English.
- (iii) The set of rivers in India.
- (iv) The set of all planets.
- (v) The set of points on a circle.
- (vi) The set of mathematics books in your library.
- (vii) The set of even positive integers (i.e., 2, 4, 6, 8, ...).
- (viii) The set of multiples of 4 (i.e., 4, 8, 12, ...).
- (ix) The set of factors of 12. (i.e., 1, 2, 3, 4, 6, 12).
- (x) The set of integers less than zero (i.e., -1, -2, -3, ...).

Notations

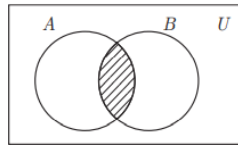
Sets are usually denoted by capital letters A, B, C , etc., and their elements by small letters a, b, c , etc. Let, A be any set of objects and let ' a ' be a member of A , then we write $a \in A$ and read it as ' a belongs to A ' or ' a is an element of A ' or ' a is a member of A '. If a is not an object of A , then we write $a \notin A$ and read it as ' a does not belong to A ' or ' a is not an element of A '.

11.3 Operation on Sets

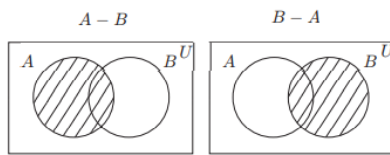
1. Union of sets: $A \cup B$



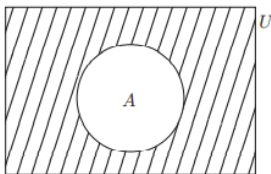
2. Intersection of sets: $A \cap B$



3. Difference of sets: $A - B$ or $B - A$



4. Complement of a set: A'



11.4 Venn Diagram with Two Attributes

In this section, we discuss the Venn diagram with two attributes and give the generalized formulae. Figure 1 shows Venn diagram representation of two sets.

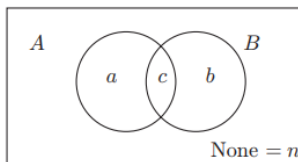
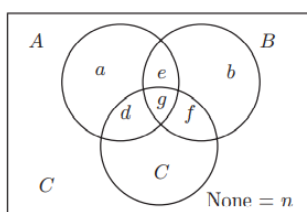


Figure 1 | Venn diagram representation of two sets A and B

1. $n(A) = a + c$
2. $n(B) = b + c$
3. Things belonging to exactly one attribute = $a + b$
4. $n(A \cap B) = c$
5. Things belonging to none of the attributes = n
6. $n(A \cup B) = a + b + c$

11.5 Venn Diagram with Three Attributes

In this section, we discuss the Venn diagram with two attributes and give the generalized formulae. Figure 2 shows Venn diagram representation of three sets.



1. $n(A) = a + e + g + d$

2. $n(B) = b + f + e + g$
3. $n(C) = c + d + g + f$
4. Things belonging to exactly one attribute = $a + b + c$
5. $n(A \cap B) = e + g$
6. $n(B \cap C) = g + f$
7. $n(A \cap C) = d + g$
8. $n(A \cap B \cap C) = g$
9. Things belonging to none of the attributes = n

11.6 Representation of Sets

There are two ways of expressing a set. These are

1. Tabular form or roster form.
2. Set-builder form or rule method.

Tabular Form or Roster Form

In this method, we list all the members of the set separating them by means of commas and enclosing them in curly brackets {}.



Example 2: Let, A be the set consisting of the numbers 1, 3, 4 and 5, then we write $A = \{1, 3, 4, 5\}$.

NOTES:- The order of writing the elements of a set is immaterial. For example, $\{1, 3, 5\}$, $\{3, 1, 5\}$, $\{5, 3, 1\}$ all denote the same set. • An element of a set is not written more than once. Thus, the set $\{1, 5, 1, 3, 4, 1, 4, 5\}$ must be written as $\{1, 3, 4, 5\}$.

11.7 Set Builder Form or rule Method

In this method, instead of listing all elements of a set, we write the set by some special property or properties satisfied by all its elements and write it as $A = \{x : P(x)\}$ or, $A = \{x \mid x \text{ has the property } P(x)\}$ and read it as “A is the set of all elements x such that x has the property P”. The symbol ‘:’ or ‘|’ stands for ‘such that’.



Example 3: Let, A be the set consisting of the elements 2, 3, 4, 5, 6, 7, 8, 9, 10. Then, the set A can be written as $A = \{x : 2 \leq x \leq 10 \text{ and } x \in \mathbb{N}\}$.

11.8 Finite and Infinite Sets

Finite Set

A set having no element or a definite number of elements is called a finite set. Thus, in a finite set, either there is nothing to be counted or the number of elements can be counted, one by one, with the counting process coming to an end.



Example 4: Each of the following sets is a finite set:

- (i) A = the set of prime numbers less than 10
= $\{2, 3, 5, 7\}$;
- (ii) B = the set of vowels in English alphabets

$= \{a, e, i, o, u\};$

(iii) $C = \{x \mid x \text{ is divisor of } 50\}.$

cardinal number of a Finite Set

The number of distinct elements in a finite set S is called the cardinal number of S and is denoted by $n(S)$.



Example 5: If $A = \{2, 4, 6, 8\}$ then $n(A) = 4$.

Infinite Set

A set having unlimited number of elements is called an infinite set. Thus, in an infinite set, if the elements are counted one by one, the counting process never comes to an end.



Example 6: Each of the following sets is an infinite set:

- (i) the set of all natural numbers $= \{1, 2, 3, 4, \dots\}.$
- (ii) the set of all prime numbers $= \{2, 3, 5, 7, \dots\}.$
- (iii) the set of all points on a given line.
- (iv) the set of all lines in a given plane.
- (v) $\{x \mid x \in \mathbb{R} \text{ and } 0 < x < 1\}.$

11.9 Empty Set or Null Set

The set which contains no element is called the empty set or the null set or void set.

The symbol for the empty set or the null set is ϕ . Thus, $\phi = \{\}$, since there is no element in the empty set.

The empty set is a finite set. Since any object x which is not equal to itself does not exist, the set $A = \{x : x \neq x\}$ is the empty set ϕ .

A set which is not empty, i.e., which has at least one element is called a non-empty set or a non-void set.



Example 7:

- (i) The set of natural numbers less than 1 is an empty set.
- (ii) the set of odd numbers divisible by 2 is a null set.
- (iii) $\{x \mid x \in \mathbb{Z} \text{ and } x^2 = 2\} = \phi$, because there is no integer whose square is 2.
- (iv) $\{x \mid x \in \mathbb{R} \text{ and } x^2 = -1\} = \phi$, because the square of a real number is never negative.
- (v) $\{x \mid x \in \mathbb{N}, 4 < x < 5\}$ is the empty set.
- (vi) $\{x \mid x \in \mathbb{Z}, -1 < x < 0\}$ is the null set.

The empty set should not be confused with the set $\{0\}$. It is the set containing one element, namely 0.

Singleton

A set containing only one element is called a singleton.



Example 8:

- (i) The set $\{0\}$ is a singleton since it has only one element 0.

Analytical Skills-I

- (ii) The set of even prime numbers is the set $\{2\}$ which is a singleton.
- (iii) $\{x \mid x \text{ is an integer and } -1 < x < 1\} = \{0\}$ is a singleton.

Equal set

Two sets A and B are said to be equal if they have the same elements and we write $A = B$. Thus, $A = B$ if every element of A is an element of B and every element of B is an element of A.

In symbols, $A = B$ iff $x \in A \Rightarrow x \in B$ and $x \in B \Rightarrow x \in A$. To indicate that two sets A and B are not equal, we will write $A \neq B$.

**Example 9:**

- (i) If $A = \{2, 3, 4\}$ and $B = \{x \mid 1 < x < 5, x \in \mathbb{N}\}$ then $A = B$.
- (ii) If $A =$ the set of letters in the word 'WOLF' and $B =$ the set of letters in the word 'FOLLOW' then $A = B$ as each = $\{W, O, L, F\}$, remembering that in a set the repetition of elements is meaningless and order of elements is immaterial.

Equivalent Sets

Two finite sets A and B are said to be equivalent if they have the same number of elements, i.e., if we can find a one-to-one correspondence between the elements of the two sets.

The symbol ' $\sim$ ' is used to denote equivalence. Thus, $A \sim B$ is read as "A is equivalent to B". Two finite sets A and B are equivalent if $n(A) = n(B)$, i.e., if they have the same cardinal number.

Equivalent sets have the same number of elements, not necessarily the same elements. The elements in two equivalent sets may or may not be the same. Thus, equal sets are always equivalent but equivalent sets may or may not be equal.

**Example 10:**

- (i) If $A = \{1, 2, 3\}$ and $B = \{2, 4, 6\}$ then $A \sim B$.
- (ii) If $A = \{a, b, c, d\}$ and $B = \{p, q, r, s\}$ then $A \sim B$.
- (iii) If $A = \{3, 5, 7, 9\}$ and $B = \{9, 7, 5, 3\}$ then $A \sim B$.

Also, since A and B have same elements, $\therefore A = B$.

Subset of set

If A and B are any two sets, then B is called a subset of A if every element of B is also an element of A.

Symbolically, we write it as $B \subseteq A$ or $A \supseteq B$

- (i) $B \subseteq A$ is read as B is contained in A or B is a subset of.
- (ii) $A \supseteq B$ is read as A contains B or A is super set of B.

**Example 11:**

- (i) The set $A = \{2, 4, 6\}$ is a subset of $B = \{1, 2, 3, 4, 5, 6\}$, since each number 2, 4 and 6 belonging to A, also belongs to B.
- (ii) The set $A = \{1, 3, 5\}$ is not a subset of $B = \{1, 2, 3, 4\}$ since $5 \in A$ but $5 \notin B$.
- (iii) The set of real numbers is a subset of the set of complex numbers. The set of rational numbers is a subset of the set of real numbers. The set of integers is a subset of the set of rational numbers. Finally, the set of natural numbers is a subset of the set of integers. Symbolically, $\mathbb{N} \subseteq \mathbb{Z} \subseteq \mathbb{Q} \subseteq \mathbb{R} \subseteq \mathbb{C}$

Proper Subsets of a Set

A set B is said to be a proper subset of the set A if every element of set B is an element of A whereas every element of A is not an element of B.

Unit 13: Logical Venn Diagram and Set Theory, Syllogism

We write it as $B \subset A$ and read it as “B is a proper subset of A”.

Thus, B is a proper subset of A if every element of B is an element of A and there is at least one element in A which is not in B.



Example 12:

- (i) If $A = \{1, 2, 5\}$ and $B = \{1, 2, 3, 4, 5\}$. Then A is a proper subset of B.
- (ii) The set N of all natural numbers is a proper subset of the set Z of all integers because every natural number is an integer, i.e., $N \subset Z$ but every integer need not be a natural number, i.e., $N \neq Z$.

Power set

Elements of a set can also be some sets. Such sets are called set of sets. For example, the set $\{f, \{1\}, \{2\}, \{3, 4\}\}$ is a set whose elements are the sets f, {1}, {2}, {3, 4}.

The set of all the subsets of a given set A is called the power set of A and is denoted by $P(A)$.



Example 13:

- (i) If $A = \{a\}$, then $P(A) = \{f, A\}$.
- (ii) If $B = \{2, 5\}$, then $P(B) = \{f, \{2\}, \{5\}, B\}$.
- (iii) If $S = \{a, b, c\}$, then $P(S) = \{f, \{a\}, \{b\}, \{c\}, \{a, b\}, \{a, c\}, \{b, c\}, S\}$.

Comparable sets

If two sets A and B are such that either $A \subset B$ or $B \subset A$, then A and B are said to be comparable sets. If neither $A \subset B$ nor $B \subset A$, then A and B are said to be non-comparable sets.



Example 14:

- (i) If $A = \{1, 3, 5\}$ and $B = \{1, 2, 3, 4, 5\}$, then A and B are comparable sets because $A \subset B$.
- (ii) If $A = B$, then A and B are comparable sets.

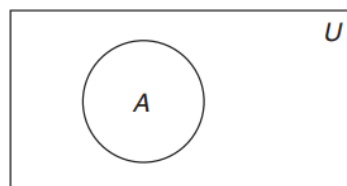
Universal sets:-

If in any discussion on set theory, all the given sets are subsets of a set U, then the set U is called the universal set. Example 15: Let, $A = \{2, 4, 6\}$, $B = \{1, 3, 5\}$, $C = \{3, 5, 7, 11\}$, $D = \{2, 4, 8, 16\}$ and $U = \{1, 2, 3, 4, 5, 6, 7, 8, 10, 11, 16\}$ be the given sets. Here the sets A, B, C, D are subsets of the set U. Hence U can be taken as the universal set.

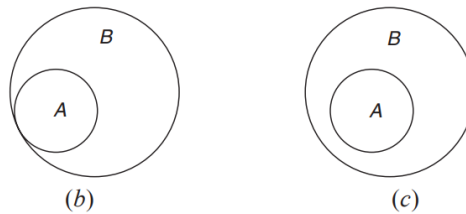
11.10 Venn Diagram

In order to visualize and illustrate any property or theorem relating to universal sets, their subsets and certain operations on sets, Venn, a British mathematician developed what are called Venn diagrams. He represented a universal set by interior of a rectangle and other sets or subsets by interiors of circles

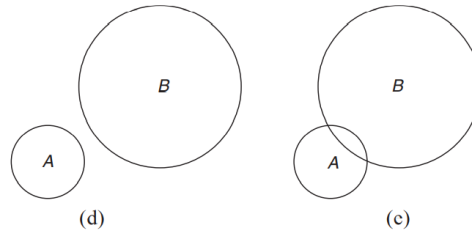
Examples of certain relationships Between Sets by Venn diagrams 1. If U be a set of letters of English alphabets and A, a set of vowels, then $A \subset U$. This relationship is illustrated by Fig. (a).



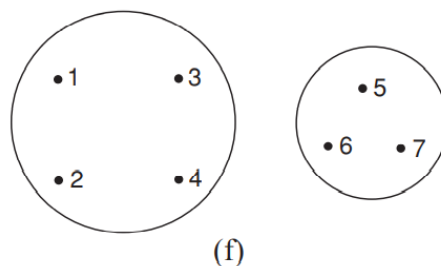
If $A \subset B$ and $A \neq B$, then A and B can be represented by either of the diagrams [Figure (b) and Figure (c)].



3. If the sets A and B are not comparable, then neither of A or B is a subset of the other. This fact can be represented by either of the diagrams [Figure (d) and Figure (e)].



4. If $A = \{1, 2, 3, 4\}$ and $B = \{5, 6, 7\}$, then A and B are disjoint. These can be illustrated by Venn diagram given in Fig. (f).

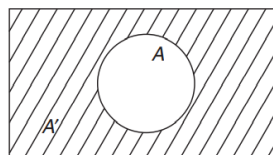


Complement of sets :-

Let, $A \subset U$ (i.e., A is a proper subset of universal set U). Evidently, U consists of all the elements of A together with some elements which are not in A. Let us now constitute another set consisting of all the elements of U not in A. Naturally, it will form another proper subset of U. We call this subset the complement of the subset A in U and denote it by A' or by A^c i.e., $A^c = \{x : x \in U, x \notin A\}$.

Thus, the complement of a given set is a set which contains all those members of the universal set that do not belong to the given set.

Example of A' by Venn diagram Let, A be a subset of the universal set U. The shaded area in figure below represents the set A' which consists of those elements of U which are not in A.



Example 16:

- (i) If the universal set is $\{a, b, c, d\}$ and $A = \{a, b, d\}$ then $A' = \{c\}$.
- (ii) If the universal set $U = \{1, 2, 3, 4, 5, 6\}$ and $A = \{2, 4, 6\}$, then $A' = \{1, 3, 5\}$.
- (iii) If $U = \mathbb{N}$ and $A = O$ (the set of odd natural numbers), then $A' = E$ (the set of even natural numbers).
- (iv) If $U = \mathbb{I}$, $A = \mathbb{N}$, then $A' = \{0, -1, -2, -3, \dots\}$.
- (v) If $U = \{1, 2, 3, 4\}$, $A = \{1, 2, 3, 4\}$, then $A' = \emptyset$.

(a) Union of Sets

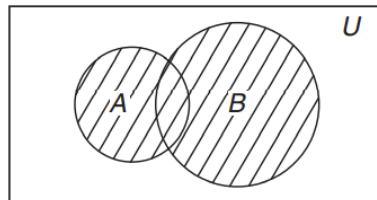
Unit 13: Logical Venn Diagram and Set Theory, Syllogism

Let A and B be two given sets. Then the union of A and B is the set of all those elements which belong to either A or B or both. The union of A and B is denoted by $A \cup B$ and is read as A union B. The symbol $\cup$ stands for union. It is evident that union is 'either, or' idea. Symbolically, $A \cup B = \{x : \text{either } x \in A \text{ or } x \in B\}$.

NOTES

the union set contains all the elements of A and B, except that the common elements of both A and B are exhibited only once. N

Example of $A \cup B$ by Venn diagram Let A and B be any two sets contained in a universal set U. Then $A \cup B$ is indicated by the shaded area in the figure below.



Example 17:

- (i) Let, $A = \{1, 2, 3, 4\}$, $B = \{2, 3, 6, 7, 9\}$, then, $A \cup B = \{1, 2, 3, 4, 6, 7, 9\}$.
- (ii) If $A = O$ (set of odd natural numbers), $B = E$ (set of even natural numbers), then $A \cup B = N$.
- (iii) If A is the set of rational numbers and B the set of irrational numbers, then $A \cup B = R$.
- (iv) If $A = \{x : x^2 = 4, x \in I\} = \{2, -2\}$, $B = \{y : y^2 = 9, y \in I\} = \{3, -3\}$, then $A \cup B = \{-3, -2, 2, 3\}$.
- (v) If $A = \{x : 1 < x < 5, x \in N\} = \{2, 3, 4\}$, $B = \{y : 3 < y < 7, y \in N\} = \{4, 5, 6\}$, then $A \cup B = \{2, 3, 4, 5, 6\}$.

NOTES

From the definition of the union of two sets A and B, it is clear that

- $x \in A \cup B \Leftrightarrow x \in A \text{ or } x \in B$
- $x \notin A \cup B \Leftrightarrow x \notin A \text{ and } x \notin B$
- $A \subseteq A \cup B$ and $B \subseteq A \cup B$.

(b) Intersection of Sets

Let A and B be two given sets. Then the intersection of A and B is the set of elements which belong to both A and B. In other words, the intersection of A and B is the set of common members of A and B. The intersection of A and B is denoted by $A \cap B$ and is read as A intersection B. The symbol $\cap$ stands for intersection. It is evident that intersection is an 'and' idea.

Symbolically,

$$A \cap B = \{x : x \in A \text{ and } x \in B\}.$$

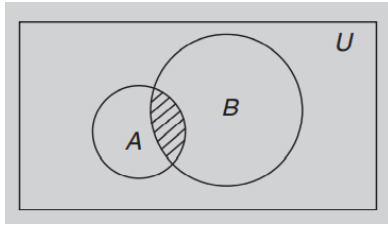
NOTES :-

From the definition of the intersection of two sets A and B, it is clear that

- $x \in A \cap B \Leftrightarrow x \in A \text{ and } x \in B$
- $x \notin A \cap B \Leftrightarrow x \notin A \text{ or } x \notin B$
- $A \cap B \subseteq A$ and $A \cap B \subseteq B$.

Let A and B be any two sets contained in the universal set U. Then $A \cap B$ is indicated by the shaded area, as shown in the figure below.

NOTES

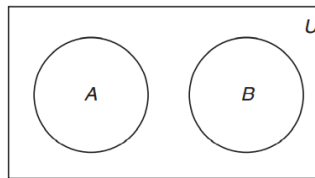


Example 18:

- (i) If $A = \{1, 2, 3, 6, 9, 18\}$, and $B = \{1, 2, 3, 4, 6, 8, 12, 24\}$, then $A \cap B = \{1, 2, 3, 6\}$.
- (ii) If A is the set of odd natural numbers and B is the set of even natural numbers, then $A \cap B = \emptyset$.
[Intersection of two disjoint sets is empty set]
- (iii) If A and B are sets of points on two distinct concentric circles, then $A \cap B = \emptyset$.
- (iv) If $A = \{x : 1 < x < 6, x \in \mathbb{N}\} = \{2, 3, 4, 5\}$, $B = \{y : 2 < y < 9, y \in \mathbb{N}\} = \{3, 4, 5, 6, 7, 8\}$
then, $A \cap B = \{3, 4, 5\}$.

Disjoint Sets

If $A \cap B = \emptyset$, then A and B are said to be disjoint sets. For example, let $A = \{2, 4, 6, 8\}$ and $B = \{1, 3, 5, 7\}$. Then, A and B are disjoint sets because there is no element which is common to both A and B . The disjoint sets can be represented by Venn diagram as shown in the figure below.



Difference of Sets

Let A and B be two given sets. The difference of sets A and B is the set of elements which are in A but not in B . It is written as $A - B$ and read as A difference B .

Symbolically,

$$A - B = \{x : x \in A \text{ and } x \notin B\}.$$

$$\text{Similarly, } B - A = \{x : x \in B \text{ and } x \notin A\}.$$

Caution: In general, $A - B \neq B - A$.

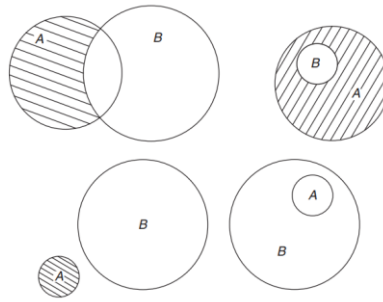


Example 19:

- (i) If $A = \{1, 2, 3, 4\}$, $B = \{2, 4, 7, 9\}$, then
 $A - B = \{1, 3\}$ and $B - A = \{7, 9\}$.
Hence $A - B \neq B - A$.
- (ii) If $A = \{12, 15, 17, 20, 21\}$,
 $B = \{12, 14, 16, 18, 21\}$
and $C = \{15, 17, 18, 22\}$, then
 $A - B = \{15, 17, 20\}$
 $B - C = \{12, 14, 16, 21\}$
 $C - A = \{18, 22\}$
 $B - A = \{14, 16, 18\}$
 $A - A = \emptyset$.

Unit 13: Logical Venn Diagram and Set Theory, Syllogism

Example of $A - B$ by Venn Diagrams In the four cases shown by the diagrams below, $A - B$ is given by shaded area.



Applications of Sets

1. If a set S has only a finite number of elements, we denote by $n(S)$ the number of elements of S .

Example 20: If $U = \{1, 2, 3, 4, 5\}$, then $n(U) = 5$.

2. For any two sets A and B , with finite number of elements, we have the following formula:

$$n(A \cup B) = n(A) + n(B) - n(A \cap B).$$

3. If A and B are disjoint sets, then $n(A \cup B) = n(A) + n(B)$.



Example 21: X and Y are two sets such that

$$n(X) = 17, n(Y) = 23, n(X \cup Y) = 38,$$

find $n(X \cap Y)$.

Solution: $n(X) = 17, n(Y) = 23, n(X \cup Y) = 38,$

$$n(X \cap Y) = ?$$

$$\text{Now, } n(X \cup Y) = n(X) + n(Y) - n(X \cap Y).$$

$$\text{Then, } 38 = 17 + 23 - n(X \cap Y)$$

$$\Rightarrow n(X \cap Y) = 17 + 23 - 38 = 2.$$

Ordered Pair

Let A and B be two non-empty sets. If $a \in A$ and $b \in B$, an element of the form (a, b) is called an ordered pair, where 'a' is regarded as 'the first element' and 'b' as the second element. It is evident from the definition that

1. $(a, b) \neq (b, a)$
2. $(a, b) = (c, d)$ if and only if $a = c$ and $b = d$.

Equality of two ordered pairs.

Two ordered pairs (a, b) and (c, d) are said to be equal if and only if $a = c$ and $b = d$. The ordered pairs $(2, 4)$ and $(2, 4)$ are equal while the ordered pairs $(2, 4)$ and $(4, 2)$ are different. The distinction between the set $\{2, 4\}$ and the ordered pair $(2, 4)$ must be noted carefully. We have $\{2, 4\} = \{4, 2\}$ but $(2, 4) \neq (4, 2)$.

CARTESIAN Product Of sets

Let, A and B be two non-empty sets. The cartesian product of A and B is denoted by $A \times B$ (read as 'A cross B') and is defined as the set of all ordered pairs (a, b) , where $a \in A$ and $b \in B$.

Symbolically, $A \times B = \{(a, b) : a \in A \text{ and } b \in B\}$



Example 22: Suppose, $A = \{2, 4, 6\}$ and $B = \{x, y\}$ Then,

$$A \times B = \{(2, x), (4, x), (6, x), (2, y), (4, y), (6, y)\}$$

Analytical Skills-I

$$B \times A = \{(x, 2), (x, 4), (x, 6), (y, 2), (y, 4), (y, 6)\}$$

Thus, we note that if $A \neq B$, then $A \times B \neq B \times A$.



Example 23: Let, $A = \{1, 2, 3\}$ and $B = \emptyset$. Then, $A \times B = \emptyset$, as there will be no ordered pair belonging

to $A \times B$. Thus, we note that $A \times B = \emptyset$ if A or B or both of A and B are empty sets.



Example 24: Let, $n(A)$ represents the number of elements in set A . In Example 22, we can see that $n(A) = 3$, $n(B) = 2$ and $n(A \times B) = 6$. Thus, we note that $n(A \times B) = n(A) \times n(B)$. In other words, if a set A has m elements and a set B has n elements, then $A \times B$ has mn elements. Further, it may be noted that $n(A \times B) = n(B \times A)$. This implies that $A \times B$ and $B \times A$ are equivalent sets.



Example 25: If there are three sets A, B, C and $a \in A, b \in B, c \in C$, we form an ordered triplet (a, b, c) . The set of all ordered triplets (a, b, c) is called the cartesian product of the sets A, B, C . That is, $A \times B \times C = \{(a, b, c) : a \in A, b \in B, c \in C\}$.

11.11 SYLLOGISMS: Deductive Reasoning

Remember: There is a difference between asserting that a premise is untrue, and asserting that the logic of the argument is faulty. "All dogs can fly. Fido is a dog. Fido can fly." That is a perfectly valid argument in terms of logic, but this flawless logic is based on an untrue premise. If a person accepts the major and minor premises of an argument, the conclusion follows undeniably by the sheer force of reason. If in an argument, the logic reaches a conclusion that seems absurd, it behooves you to analyze each sentence separately (to see if each premise is true without exception) and then to analyze the structure of the argument (to see if the reasoning of the argument itself is valid). Also be on the lookout for "equivocation," the use of two different meanings of one word during the process of an argument.

Directions: Decide whether the following syllogisms are valid in format (as opposed to using true premises). If one is invalid, either explain why, or illustrate using circles.

Invalid example: All snakes are cold-blooded.

All snails are cold-blooded.

All snails are snakes.

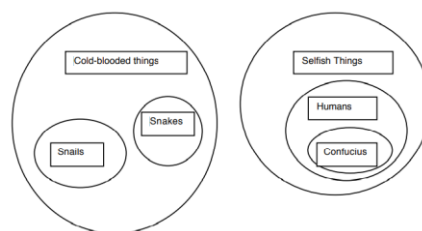
INVALID: Snakes and snails could overlap as categories, but they both could also premises are true, but if we accept these be in the largest circle without overlapping.

VALID. We might not agree that the individual premises are true, but if we accept these premises, the conclusion does logically follow.

Valid example: All humans are selfish.

Confucius is a human.

Confucius is selfish.



1) All human societies are doomed to deteriorate over time.

America is a human society.

America is doomed to deteriorate over time.

Unit 13: Logical Venn Diagram and Set Theory, Syllogism

2) No philosophers are evil.

All Greeks are philosophers.

No Greeks are evil.

3) All women are potential mothers.

Betty is a potential mother.

Betty is a woman.

4. All students are eligible for student government.

No teachers are eligible for student government.

No teachers are students.

5. All barbiturates are drugs.

Marijuana is not a barbiturate.

Marijuana is not a drug.

6. Pizza is a substance made of cardboard.

All substances made of cardboard are good to eat.

Pizza is a substance that is good to eat.



Example 26

Statements Some cameras are radios

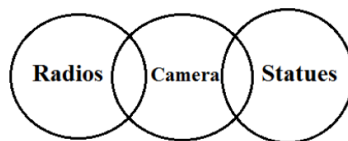
Some statues are cameras.

Conclusions I. Some radios are statues.

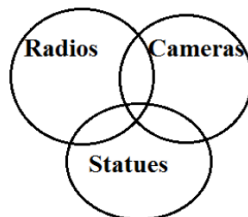
II. No radio is statue.

Solution. Either “some radios are statues” or “no radio is statue”, as I and E-type propositions form a complementary pair.

We can draw all possible cases as given below



b) Hence using both diagrammatic representation, we can conclude either ‘some radios are statues’ or ‘no radio is statue.’ Hence, atleast, one of the conclusions must be true.



Special Cases

Facts

Combination (Conclusion) Conclusion

Analytical Skills-I

A + O

Either I or II follows

E + I

Either I or II follows

I + O

Either I or II follows



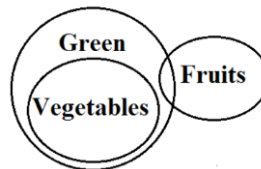
Example 27. Statements All vegetables are green.

Some greens are fruits.

Conclusions I. Some fruits are vegetables.

II. No fruit is vegetable.

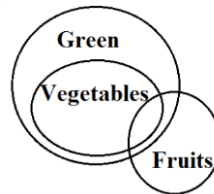
Solution. Here, Conclusion I is particular affirmative and Conclusion II is universal negative proposition. Hence, either Conclusion I or Conclusion II follows.



(b)

Conclusion If we follow venn diagram (a), then we can say no fruit is vegetable Conclusion II but if we follow venn diagram

(b), then we can say some fruits are vegetables. (Conclusion I) Here either venn diagram (a) or venn diagram (b) is possible. Hence, Conclusion I or Conclusion II must be followed. Minimal Possibilities We can represent statements by keeping in mind one conclusions. If we follow that our two conclusions belong to special case, then either one of them is true.



Example 28.

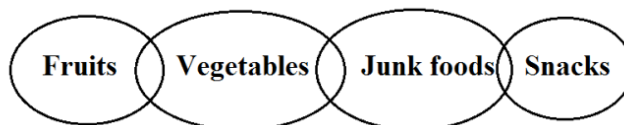
Statements Some fruits are vegetables.

Some vegetables are junk food.

Conclusions I. Some junk foods are vegetables.

II. Some junk food are fruits.

Solution.



It is clear the above diagram that only Conclusion I follows.

Possibility.

Possibility is a concept of inconsistency for an event which is not yet verified but if true would explain certain facts or phenomena. In other words capability of existing or happening or being true known as possibility

Unit 13: Logical Venn Diagram and Set Theory, Syllogism

Condition

Possibility

Given facts Imaginary facts

Cannot be determined Can be determined

It is so simple to understand through the above table that possibility exists where no definite relation occurs between the objects and definite relation between the objects eliminate existence of any possibility. In simple way given with an example which will also clear the term possibility.



Example 29

Statements Some birds are trees.

Some trees are hens.

Conclusions I. Some birds being hens is a possibility.

II. All trees being hens is a possibility.

Solution. In Conclusion I, before deciding the possibility between birds and hens, we must notice the relation between both, we find that there is no relation between birds and hens, so possibility favours the condition and the Conclusion I is true for possibility and in Conclusion II we must notice the relation between trees and hens. We find that both have some type of relation between them so the possibility of 'All' between trees and hens is true. Hence, both the Conclusions I and II follows.

Given Exclusive Proportion	Desired Proportions	Possibility
All	All	No
Some	Some	No
No	No	No
No	Some not	No
Some	All	Yes
No proper relations	Some All	Yes

Note Improper relation between two objects favours the possibility (In above example Conclusion I)

Special Cases of Exclusive Proposition

If the statement is of	Conversion	Example	Meaningful Conversion
Much, more, many, very, a few, most, almost Atleast Definitely Only 1% to 99%	Some Some No use Some	Most A are B. A few X are Y. Atleast some A are B. Some A are definitely B. Some X are definitely not Y. Only A are B. 38% A are B.	Some A are B. Some X or Y. Some A are B. Some A are B. Some X are not Y. All B are A. Some A are B. Some X are Y.

		98% X are Y.	
--	--	--------------	--

Summary

The key concepts learned from this unit are: -

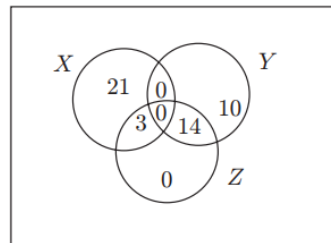
- We have learnt how to show the relationships among things or finite groups of things.
- We have learnt what are sets
- We have learnt what are different types of sets
- We have learnt what are operations on a sets
- We have learnt what is Syllogism.

Keywords

- Logical Venn diagram
- set theory
- Syllogism

Self Assessment

Direction (Q1–Q5): Solve the examples on the basis of the following diagram:



Note: X represents even numbers, Y represents odd numbers and Z represents prime numbers in the sample.

1. How many even numbers are there which are not prime?
A. 21
B. 22
C. 24
D. 45
2. How many odd numbers are there which are prime?
A. 21
B. 14
C. 25
D. 49
3. How many even numbers are prime?
A. 1
B. 12
C. 3
D. 4

Unit 13: Logical Venn Diagram and Set Theory, Syllogism

4. How many numbers are present which are not prime?

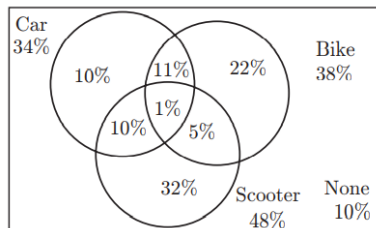
- A. 31
- B. 12
- C. 32
- D. 42

5. How many total odd numbers are present?

- A. 30
- B. 12
- C. 33
- D. 24

Direction (Q6–Q10): In a survey conducted for a society, 10% of the houses have just car, 22% have

just bike, 32% have just scooter, 1% have a car, bike and scooter, and 10% have none of the three things. So, 40 houses do not have any vehicle.



Say x is $(\text{car} \cap \text{bike})$, y is $(\text{car} \cap \text{scooter})$ and z is $(\text{bike} \cap \text{scooter})$. Therefore,

$$31 = 10 + x + 1$$

$$38 = 22 + x + 1 + z$$

$$48 = 32 + y + 1 + z$$

Solving the above equations for x , y and z , we get

$$x = 10\%$$

$$y = 10\%$$

$$z = 5\%$$

$$\text{Also, total houses surveyed} = \frac{100 \times 40}{10} = 400$$

6. How many houses have bikes only?

- A. 35
- B. 10
- C. 33
- D. 88

7. How many houses have exactly two vehicles? Solution: Total houses with more than one vehicle = 25%

- A. 36
- B. 96
- C. 39

D. 80

8. How many houses have only cars?

A. 30

B. 12

C. 39

D. 96

9. How many houses have only one vehicle?

A. 35

B. 100

C. 256

D. 240

10. How many houses do not have a scooter?

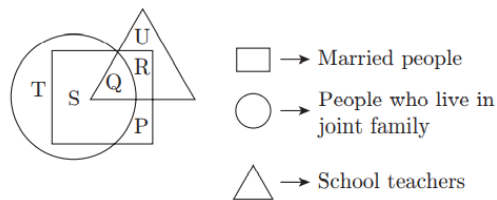
A. 168

B. 102

C. 340

D. 812

Direction (Q11 – Q13): The following Venn diagram depicts people living in a building who are married, who live in joint family and who are school teacher. Study the chart carefully and answer the following.



11. By which letter, the married teachers who live in joint family are represented?

A. U

B. R

C. P

D. Q

12. By which letter, the married people who live in joint family but not are school teachers are represented?

A. S

B. U

C. R

D. P

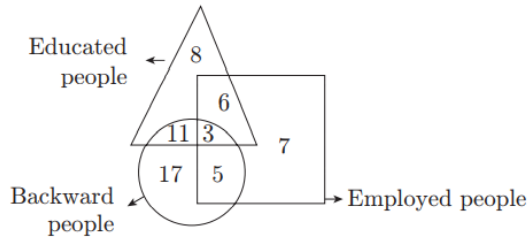
13. The people who live in joint family but are neither married nor teachers are represented by which letter?

A. S

B. U

- C. R
D. T

Direction (Q14–Q15): The following Venn diagram depicts people who are educated, backward and employed. Study the diagram carefully and answer the following.



14. How many educated people are employed?

- A. 9
B. 10
C. 12
D. 15

15. How many backward people are educated?

- A. 32
B. 11
C. 14
D. 50.

16. Statements:

Some ropes are walls.

Some walls are sticks.

All sticks are chairs.

All chairs are tables.

Conclusions:

I. Some tables are walls.

II. Some chairs are ropes.

III. Some sticks are ropes.

- A. None follows
B. Only I follows
C. Only II follows
D. Only III follows

17. Statements:

Some pens are knives.

All knives are pins.

Some pins are needles.

Analytical Skills-I

All needles are chains.

Conclusions:

I. Some chains are pins.

II. Some needles are knives.

III. Some pins are pens.

A. Only I follows

B. Only II follows

C. Only III follows

D. None of these

Answer for Self Assessment

- | | | | | |
|-------|-------|-------|-------|-------|
| 1. A | 2. B | 3. C | 4. A | 5. D |
| 6. D | 7. B | 8. D | 9. C | 10. A |
| 11. D | 12. A | 13. D | 14. A | 15. C |
| 16. B | 17. D | | | |

Review Questions

1. Specify the set A by listing its elements, where

$A = \{ \text{whole numbers less than 100 divisible by 16} \}$.

b Specify the set B by giving a written description of its elements, where

$B = \{ 0, 1, 4, 9, 16, 25 \}$.

c Does the following sentence specify a set?

$C = \{ \text{whole numbers close to 50} \}$.

$C = \{ \text{whole numbers close to 50} \}$.

2. Use dots to help list each set, and state whether it is finite or infinite. i $A = \{ \text{even numbers between 10 000 and 20 000} \}$ ii $B = \{ \text{whole numbers that are multiples of 3} \}$
3. If the set S in each part is finite, write down $|S|$. i $S = \{ \text{primes} \}$ ii $S = \{ \text{even primes} \}$ iii $S = \{ \text{even primes greater than 5} \}$ iv $S = \{ \text{whole numbers less than 100} \}$
4. Let F be the set of fractions in simplest form between 0 and 1 that can be written with a single-digit denominator. Find F and $|F|$.
5. Rewrite in set notation:
 - i All squares are rectangles.
 - ii Not all rectangles are rhombuses.
6. Rewrite in an English sentence using the words 'all' or 'not all':
 - i $\{ \text{whole number multiples of 6} \} \subseteq \{ \text{even whole numbers} \}$.
 - ii $\{ \text{square whole numbers} \} \subseteq \{ \text{even whole numbers} \}$.
7. Rewrite the statements in part (6) in an English sentence using the words 'if ..., then'. d
Given the sets $A = \{ 0, 1, 4, 5 \}$ and $B = \{ 1, 4 \}$:
 - i Draw a Venn diagram of A and B using the universal set $U = \{ 0, 1, 2, \dots, 8 \}$.

Unit 13: Logical Venn Diagram and Set Theory, Syllogism

- ii Graph A on the number line.
8. Let the universal set be $E = \{\text{whole numbers less than } 20\}$, and let $A = \{\text{squares less than } 20\}$ $B = \{\text{even numbers less than } 20\}$ $C = \{\text{odd squares less than } 20\}$
 - i) Draw A and C on a Venn diagram, and place the numbers in the correct regions.
 - ii) Draw B and C on a Venn diagram, and place the numbers in the correct regions.
 - iii) Shade $A \cap B$ on a Venn diagram, and place the numbers in the correct regions.
 - iv) Shade $A \cup B$ on a Venn diagram, and place the numbers in the correct regions.
 9. Draw a Venn diagram of two sets S and T b Given that $|S| = 15$, $|T| = 20$, $|S \cup T| = 25$ and $|E| = 50$, insert the number of elements into each of the four regions. c Hence find $|S \cap T|$ and $|S \cap T^c|$.
 10. Twenty-four people go on holidays. If 15 go swimming, 12 go fishing, and 6 do neither, how many go swimming and fishing? Draw a Venn diagram and fill in the number of people in all four regions.
 11. In a certain school, there are 180 pupils in Year 7. One hundred and ten pupils study French, 88 study German and 65 study Indonesian. Forty pupils study both French and German, 38 study German and Indonesian only. Find the number of pupils who study: a all three languages b Indonesian only c none of the languages d at least one language e either one or two of the three languages



Further Readings

1. Quantitative Aptitude For Competitive Examinations By Dr. R S Aggarwal, S Chand Publishing
2. A Modern Approach To Verbal & Non-Verbal Reasoning By Dr. R S Aggarwal, S Chand Publishing
3. Magical Book On Quicker Maths By M Tyra, Banking Service Chronicle
4. Analytical Reasoning By M.K. Pandey, Banking Service Chronicle